

Exact Belief-State Computation at Scale II: The Four-Parameter Cube and the Antichain Census

A computational companion to the belief-state safety-value theory

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Abstract

Edition 1 of this series scaled the belief-state safety value to a medium instance of 48 augmented cells and argued the antichain as the stored object. This edition pushes the same exact discipline into the curse-of-dimensionality regime: a four-parameter cube with $2^4 = 16$ hidden-parameter cells on 16 stock levels — 256 augmented cells, 17 controls, every computation in exact integer and rational arithmetic. The structure is classified completely. A pairwise-Hamming lemma (every pair of distinct actions sums, over any two parameter cells, to at most twice their agreement count) forces every blind survivable set to consist of Hamming-adjacent cells, and no three cells of the four-cube are pairwise adjacent: no blind policy of any period keeps three or more cells safe. The bands are exact: 16 maximal singletons at the floor's edge, exactly the 32 Hamming-adjacent pairs at every level above it (the matched-singly-mismatched alternation rises at net $\frac{1}{5}$ per two steps with a $\frac{1}{10}$ dip). The observation ladder doubles: each additional exact probe of one parameter doubles the conditional kernel mass — $\frac{1}{16} \rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1$ at the edge, $\frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1$ above it — so three probes exhaust observation above the edge. Exact point-based evaluation is verified against definition-level enumeration at seven rational beliefs, three levels, four horizons; the four-step alpha-set deduplicates 83,521 action sequences to at most 545 distinct outcome vectors. The antichain census is the headline: 1,048,576 raw subset evaluations collapse to 496 stored maximal sets. A third instance of the companion theory's additive deadline law closes the edition (threshold $1 + T/2$, zero probe excursion).

Keywords: exact computation; partially observable control; zero-suppressed sets; survivable sets; Hamming structure; observation design

0.1 Introduction

Edition 1 established the two working disciplines — evaluate exactly where you probe, store the antichain not the lattice — on a two-parameter instance of 48 augmented cells, where hand analysis had failed three times and enumeration was the arbiter. This edition asks the natural next question: does the exact discipline survive the curse of dimensionality, and what new structure appears there?

The answer is yes, with a surprise: the four-parameter cube is not a bigger version of the two-parameter instance but a different animal. Its survivable sets are governed by Hamming adjacency rather than by coordinate matching; its observation ladder is geometric rather than one-shot; and its raw subset lattice (2^{16} per level) is four orders of magnitude larger than its stored antichain. All claims below are regenerated by the edition's verification script in exact integer and rational arithmetic, standard library only, which chains edition 1's script as its seed.

0.2 The cube instance

The state is (z, θ) with stock z on the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and hidden parameter $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \{+1, -1\}^4$: 16 parameter cells on 16 levels, 256 augmented cells. Controls are $u \in \{+1, -1\}^4 \cup \{(0, 0, 0, 0)\}$ (17 actions), and the one-step drift is

$$\dot{z} = d(u, \theta) = -\frac{1}{2} + \frac{1}{5} \sum_{i=1}^4 u_i \theta_i :$$

the matched action $u = \theta$ rises at $+\frac{3}{10}$; a single mismatch gives $-\frac{1}{10}$, two give $-\frac{1}{2}$, three give $-\frac{9}{10}$, the antimatched action $-\frac{13}{10}$, and the hold action $-\frac{1}{2}$ for every cell. The floor is $z \geq 1$. The sum over all sixteen cells of any action's drifts is -8 per step: the cube as a whole always declines, and the blind controller declared here sees no observation of θ .

0.3 The pairwise-Hamming classification

Lemma 1 (pair-sum bound). *For any two parameter cells θ, θ' at Hamming distance h and any action $u \in \{+1, -1\}^4$,*

$$\langle u, \theta \rangle + \langle u, \theta' \rangle \leq 2(4 - h),$$

and the hold action gives 0 for every pair. A blind policy that keeps both cells above the floor indefinitely forces each cell's time-averaged inner product to be at least $\frac{5}{2}$ (the drift is $-\frac{1}{2} + \frac{1}{5}\langle u, \theta \rangle$, nonnegative on average exactly at $\frac{5}{2}$), so the pair average is at least $5 \leq 2(4 - h)$: $h \leq 1$.

Lemma 2 (no adjacency triangles). *No three distinct cells of $\{+1, -1\}^4$ are pairwise at Hamming distance 1. Hence every blind survivable set has at most two members, under*

every blind policy — any period, and no period at all. The bound of Lemma 1 is verified for all 120 cell pairs and all 17 actions, and Lemma 2 over all $\binom{16}{3} = 560$ triples, by the edition’s script.

Proposition 3 (bands and maximality). *A singleton $\{\theta\}$ is survivable from every $z_0 \geq 1$ (matched play rises at $+\frac{3}{10}$). A Hamming-adjacent pair $\{\theta, \theta'\}$ is survivable exactly from $z_0 \geq 1.1$: the alternation (θ, θ') gives each member drifts $+\frac{3}{10}, -\frac{1}{10}$ per two steps — net $+\frac{1}{5}$ with a within-cycle dip of $\frac{1}{10}$ — certified at 60 steps, and from $z_0 = 1.0$ the first cell to act under its mismatched action dips below the floor. The maximal survivable sets are therefore exactly the 16 singletons at $z_0 = 1.0$ and exactly the 32 Hamming-adjacent pairs at every $z_0 \geq 1.1$, and nothing larger under any blind policy. An exhaustive search over all 5,219 one-, two-, and three-periodic policies with 60-step certificates reproduces this classification independently.*

Remark 4 (the crude instrument is not sufficient). The averaging identity used on smaller instances — a survivable set S needs $\sum_i |\sum_{\theta \in S} \theta_i| \geq \frac{5}{2} |S|$ — is not sufficient here: the four-set $\{++++, +++-, ++-+, +--+\}$ attains the bound with equality ($4 + 2 + 2 + 2 = 10$) yet is not survivable, since it pairs cells at Hamming distance 2. The pairwise instrument of Lemma 1 is the sharp one.

0.4 The observation ladder doubles

Proposition 5 (geometric ladder). *Prior b_0 is uniform on the sixteen cells; a probe of one parameter refines the support to a subcube. The maximal conditional kernel mass doubles with each probe:*

$$\begin{aligned} \frac{1}{16} &\rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \quad (z_0 = 1.0), \\ \frac{1}{8} &\rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1 \quad (z_0 \geq 1.1), \end{aligned}$$

for zero, one, two, three, and four probes. Above the band’s edge three probes exhaust observation — the support is then a Hamming-adjacent pair, jointly survivable, and the fourth probe buys nothing — while at the edge the fourth probe is worth $\frac{1}{2}$, the single-cell mass the pair’s dip excludes. All ten values are computed exactly, by survivable-subset enumeration within each support.

0.5 Exact point-based evaluation at four parameters

Proposition 6 (exact alpha-set values). *The alpha-vector representation of the companion theory (its class-restricted rational alpha-vectors) evaluated at probed beliefs returns the exact value: for horizons $k = 1, 2, 3, 4$, three stock levels, and seven random rational beliefs, the maximum over the deduplicated alpha-set equals the maximum over definition-level trajectory enumeration — all 84 pairings equal. The four-step alpha-set deduplicates $17^4 = 83,521$ action sequences to at most 545 distinct outcome vectors per level (and to a single vector at the top level’s one-step horizon, where every action is safe — the silent region of the cube). Point-based selection chooses where to evaluate, never what the value is.*

0.6 The antichain census

Proposition 7 (the antichain, not the lattice). *The raw subset lattice carries $2^{16} = 65,536$ subsets per level —*

$16 \times 65,536 = 1,048,576$ over the grid — while the stored object is the antichain of maximal survivable sets: 16 sets at the edge and 32 per level above it, 496 stored sets in total. The per-level compression ratio is $4,096\times$ at the edge and $2,048\times$ above it; and by Lemma 2 nothing is lost — the antichain is the whole classification, not a summarization of one.

0.7 A deadline instance

Proposition 8 (the deadline law on the cube). *Hold for T steps (drift $-\frac{1}{2}$), let the parameter be revealed exactly at T , then play matched forever (drift $+\frac{3}{10}$ on every branch). The companion theory’s additive deadline law applies with blind-drift cost $d_0 = \frac{1}{2}$ and zero probe excursion: viability holds exactly when*

$$z_0 \geq 1 + \frac{T}{2}, \quad T = 0, \dots, 4,$$

verified by direct simulation on the grid. The excursion vanishes because the learned act rises on every branch immediately — the law’s e_p term prices the probe step the free revelation does not need.

0.8 Verification methods

Every number is regenerated by one deterministic script in exact integer and rational arithmetic, standard library only, which first chains edition 1’s script (all eleven of its checks) as the seed: the pair-sum bound over all 120 pairs and 17 actions; the adjacency-triangle check over all 560 triples; the exhaustive one-, two-, three-periodic classification with 60-step certificates; the ten ladder values by survivable-subset enumeration within each support; the 84 point-based pairings against definition-level enumeration; the census counts; the deadline thresholds by direct simulation; and every headline number asserted present in this text.

0.9 Delimitations

The blind class is declared and load-bearing: the policy may not depend on the belief, so a single action sequence governs every parameter branch and the pair-sum argument applies; a z -feedback policy could branch its actions across branches (the branches’ stock paths diverge) and is outside the classification, as in edition 1 and the companion theory’s class discipline. Drifts are deterministic per branch; probes are exact and cost no stock beyond the blind decline they interrupt. Whether some non-blind discipline beats the pairs on the cube is not claimed either way. Scaling beyond sixteen cells — where even the antichain’s input enumeration becomes the bottleneck — is the next step, for which the pairwise-Hamming instrument is the argued generalization target.

0.10 Conclusion

The exact discipline survives the curse of dimensionality and the cube repays it with structure a pencil would never find: Hamming governance of survivable sets, a doubling observation ladder, a silent region at the top of the stock grid, and a four-order antichain census. The stored object is the antichain; the argued instrument is the pairwise

bound; and the two together are what make 1,048,576 subset evaluations collapse to 496 sets without loss.

References

Abaee, A. An obstruction calculus for viability under incomplete observation (submitted).

Abaee, A. Probabilistic sufficiency for the obstruction calculus: belief-state safety values under partial observation (companion theory paper, same repository).

Abaee, A. Exact belief-state computation at scale: point-based recursion and survivable-set compression (edition 1 of this series, same repository).

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_exact_belief_computation_v2_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it chains edition 1's script and reproduces all bounds, bands, ladder values, pairings, census counts, and thresholds verbatim.

AI declaration

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